

Notes: Friberg, Goldstein, and Hankins (JFE, 2024)

Corporate Responses to Stock Price Fragility

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1 Big picture

- **Question.** Do firms change real and financial policies when their shareholder base makes their stock price more fragile to non-fundamental demand shocks?
- **Core idea.** Stock price fragility is a forward-looking financing risk. If future equity prices may be temporarily underpriced, a firm facing a cash shortfall may have to raise costly external finance.
- **Main mechanism.** A higher likelihood of future misvaluation raises the marginal benefit of precautionary cash. The firm holds more cash today and cuts long-term investment or other liquidity uses.
- **Empirical measure.** The paper uses the Greenwood and Thesmar stock price fragility measure, which combines ownership concentration and correlated mutual-fund flows.
- **Main panel result.** Within-firm changes in fragility are positively related to cash holdings. The benchmark cash coefficient on $\sqrt{\text{Fragility}}$ is positive and statistically significant.
- **Future misvaluation evidence.** Increases in fragility predict future undervaluation and overvaluation indicators, which supports the interpretation of fragility as misvaluation risk.
- **Heterogeneity.** The cash response is stronger when uncertainty is high, when firms are financially constrained, when firms are small, when they lack bond ratings, and when analyst coverage is low.
- **Current misvaluation controls.** The cash result survives controlling for many current misvaluation proxies, including book-to-market, residual valuation measures, mutual-fund outflows, and future excess returns.
- **Other policies.** Fragility is associated with lower capital expenditures, lower R&D, lower repurchases, and lower short-term debt, but little evidence of a change in capital structure.
- **Natural experiment.** The BlackRock–BGI merger creates an exogenous ownership-concentration shock. Treated firms increase cash by about 1.3 percentage points and reduce investment/liquidity payouts.
- **Alternative shocks.** Bank of America–Fleet and JP Morgan–Bank One mergers provide additional evidence that ownership shocks raise cash, although effects are smaller.
- **Economic takeaway.** Managers treat stock price fragility as a salient corporate risk. Non-fundamental stock-market shocks matter not only after mispricing occurs, but also ex ante through precautionary corporate policies.

2 Data, variables, and measurement

2.1 Sample and dependent variables

- The sample uses quarterly Compustat data from 2001 Q1 through 2017 Q4.
- The sample excludes utilities, financial firms, and SIC 9000 firms.
- Firms must have positive book equity and positive sales.
- Main dependent variables are scaled by total assets:

$$\text{Cash}, \quad \text{CapEx}, \quad \text{R\&D}, \quad \text{Repurchases}, \quad \text{Dividends}, \quad \text{STDebt}.$$

- Capital-structure outcomes include equity issuance, long-term debt, and book leverage.

Interpretation: The sample is designed to study operating firms whose cash, investment, and financing decisions can respond to stock-market fragility.

2.2 Stock price fragility

The paper follows Greenwood and Thesmar and measures firm-level fragility as

$$G_{it} = \left(\frac{1}{\theta_{it}} \right)^2 W_{it}' \Omega_t W_{it}, \quad (1)$$

where W_{it} is the vector of mutual-fund ownership weights in stock i , Ω_t is the covariance matrix of mutual-fund flows, and θ_{it} is the market capitalization of the firm.

Monthly percentage flows for fund j are

$$f_{jt}^{\%} = \frac{TNA_{jt} - TNA_{j,t-1}(1 + R_{jt})}{TNA_{j,t-1}}, \quad (2)$$

and the dollar-flow covariance matrix is transformed as

$$\hat{\Omega}_t = \text{diag}(TNA_t) \Omega_t^{\%} \text{diag}(TNA_t). \quad (3)$$

Interpretation: Fragility is high when ownership is concentrated in funds whose flows are correlated. Such a firm is more exposed to non-fundamental demand shocks from investor flows.

2.3 Controls

The baseline controls include:

- earnings volatility;
- log assets;
- institutional ownership;
- leverage;
- operating cash flow;
- fixed assets;
- inventory.

Interpretation: The empirical design asks whether changes in ownership-induced fragility matter after controlling for standard determinants of cash and investment.

2.4 Current misvaluation proxies

To separate expected future misvaluation from current valuation, the paper controls for:

- Stambaugh-Yuan-Yu mispricing score;
- book-to-market;
- residual book-to-market measures from Pástor-Veronesi, Hoberg-Phillips, and Rhodes-Kropf-Robinson-Viswanathan;
- mutual-fund outflows;
- future excess returns.

Interpretation: These controls reduce the concern that fragility simply captures current overvaluation or undervaluation rather than expected future price instability.

3 Model and empirical specifications

3.1 Precautionary cash model

The firm chooses a cash buffer x and gives up return from a long-term project, denoted by $h(x)$. Mispricing is Δp :

$$\Delta p > 0 \quad \text{overpricing}, \quad \Delta p < 0 \quad \text{underpricing}.$$

Let Δp^+ denote positive realizations and Δp^- denote the absolute value of negative realizations. Financing gains when overvalued are $G(\Delta c, \Delta p^+)$ and financing losses when undervalued are $L(\Delta c, \Delta p^-)$.

The first-order condition is

$$h'(x) = 1 + \int_{\underline{e}}^{c^* - x} \left[\int_{-\infty}^0 L_1(c^* - x - e, \Delta p^-) g(\Delta p) d\Delta p \right] f(e) de. \quad (4)$$

Interpretation: The marginal cost of a larger cash buffer is foregone investment return. The marginal benefit is avoiding future financing losses when cash is short and the stock is underpriced. Overpricing gains do not enter the cash-buffer choice because the firm can exploit overpricing regardless of its cash level.

3.2 Main panel specification

For firm i in quarter t , the main regression is

$$\frac{\text{DEP}_{it}}{\text{Assets}_{it}} = \phi_i + \gamma_{qt-ind} + \alpha \sqrt{\text{Fragility}_{it-1}} + \beta X_{it-1} + \varepsilon_{it}. \quad (5)$$

- DEP is cash, investment, liquidity management, or financing policy.
- ϕ_i are firm fixed effects.
- γ_{qt-ind} are quarter-by-industry fixed effects.
- X is the vector of lagged controls.
- Standard errors are clustered at the firm level.

Interpretation: The coefficient α measures whether within-firm increases in fragility are followed by changes in corporate policy.

3.3 Heterogeneity specification

The paper estimates interactions of fragility with high-risk or high-constraint indicators:

$$\frac{\text{DEP}_{it}}{\text{Assets}_{it}} = \phi_i + \gamma_{qt-ind} + \alpha_H \sqrt{\text{Fragility}_{it-1}} \times \text{High}_{it-1} + \alpha_F \sqrt{\text{Fragility}_{it-1}} + \theta \text{High}_{it-1} + \beta X_{it-1} + \xi_{it}. \quad (6)$$

Interpretation: The interaction coefficient tests whether fragility matters more when the cost of future misvaluation is higher.

3.4 Future misvaluation tests

The paper estimates OLS and logit regressions of future undervaluation and overvaluation indicators on increases in fragility at 2-, 4-, and 6-quarter horizons.

Interpretation: If fragility is a measure of exposure to future misvaluation, it should predict both future underpricing and future overpricing, not just current price levels.

3.5 Natural-experiment design

The main shock is the 2009 BlackRock–BGI merger. The treatment indicator equals one for firms jointly held by BlackRock and BGI before the merger and zero for firms held by only one of the two institutions:

$$Y_{it} = \phi_i + \gamma_{qt-ind} + \delta MergerTreatment_{it} + \beta X_{it-1} + u_{it}. \quad (7)$$

Interpretation: Because pre-merger ownership was already set and the merger was externally driven, the event provides plausibly exogenous variation in ownership concentration and fragility.

4 Identification and empirical design

4.1 What identifies the panel effects

- Identification comes from within-firm variation in stock price fragility.
- Firm fixed effects absorb time-invariant firm characteristics.
- Quarter-industry fixed effects absorb common industry-time shocks.
- Lagged controls address time-varying operating and financing characteristics.

Interpretation: The identifying assumption is that, conditional on fixed effects and controls, changes in mutual-fund ownership-induced fragility are not driven by unobserved changes in the firm’s desired cash and investment policies.

4.2 Why future misvaluation tests matter

- The model predicts that managers react to expected future misvaluation risk.
- Table 2 shows that increases in fragility are associated with both future undervaluation and overvaluation.
- This supports the idea that fragility captures price instability, rather than simply a directional overpricing or underpricing signal.

Interpretation: The future misvaluation tests connect the ownership-based fragility measure to the model’s financial risk channel.

4.3 Why the BlackRock–BGI merger matters

- The merger was large and unexpected.
- It changed ownership concentration for firms jointly held by BlackRock and BGI.
- Pre-trends in cash and fragility are not evident in the event-study figure.
- The treatment increases return volatility and excess-return volatility, consistent with increased fragility.

Interpretation: The merger evidence supports a causal interpretation of the panel regressions: firms respond to ownership shocks that raise fragility.

4.4 What the paper cannot fully prove

- The panel regressions are observational and may still contain residual endogeneity.
- Fragility may correlate with other aspects of investor clientele or governance.
- The model abstracts from many details of capital raising, debt markets, and real investment technology.

- The merger designs are stronger than the panel regressions, but treatment may still interact with macro conditions and product-market features.
- The paper focuses on U.S. public firms, so the external validity to private firms or non-U.S. markets is not automatic.

Interpretation: The empirical case is strongest because the same pattern appears in the panel, in heterogeneity tests, in current-misvaluation controls, and in multiple ownership-shock designs.

5 Tables and figures

This section keeps the compact visual style used in the prior notes. Adjacent original tables and figures are grouped together to save space and improve reading speed. All visuals below are cropped directly from the paper.

5.1 Table 1 + Table 2: sample, variables, and future misvaluation

Read: Table 1.

- Average cash is 19.2% of assets; median cash is 10.5%.
- Average capital expenditure is 1.2% of assets and average R&D is 2.4% of assets.
- Mean institutional ownership is 52.2%.
- The mean of $\sqrt{\text{Fragility}}$ is 0.004.

Read: Table 2.

- Increases in fragility predict future undervaluation at 2-, 4-, and 6-quarter horizons.
- They also predict future overvaluation at the same horizons.
- Both OLS and logit estimates support the interpretation of fragility as future misvaluation exposure.

Economic message: The first table sets the scale of the sample and variables. The second table validates the key interpretation: fragility captures future price instability, not merely current valuation.

Table 1

Summary Statistics.

	Mean	Sd	P50	N
Cash/Assets	0.192	0.218	0.105	136,191
CapEx/Assets	0.012	0.017	0.007	135,833
R&D/Assets	0.024	0.034	0.014	73,532
Repurchases/Assets	0.005	0.013	0.000	127,111
Dividends/Assets	0.002	0.005	0.000	135,585
ST Debt/Assets	0.040	0.077	0.008	135,255
$\sqrt{\text{Fragility}}$	0.004	0.004	0.003	136,191
Earnings volatility	0.025	0.048	0.010	136,191
Ln(Assets)	6.176	2.000	6.121	136,191
Inst Ownership	0.522	0.350	0.612	136,191
Leverage	0.196	0.207	0.135	136,191
Fixed Assets	0.246	0.224	0.170	136,191
Inventory	0.124	0.138	0.084	136,191
Oper Cash Flow	0.010	0.051	0.020	136,191

Summary statistics for variables as used in baseline regressions: Greenwood and Thesmar's stock price fragility measure as well as additional firm-level control variables. The data is quarterly from 2001 - 2017 and the sample excludes utilities, financial firms, and SIC 9000 codes. We require positive book equity and positive sales. Variables that are reported as year-to-date are transformed to quarterly flow variables based on the fiscal year-end and generally scaled

Table 1: summary statistics

Table 2
Increase in Stock Fragility and Future Misvaluation.

Panel A	Future Undervaluation Indicator			Logit		
	OLS					
	(1) +2 qtr	(2) +4 qtr	(3) +6 qtr	(4) +2 qtr	(5) +4 qtr	(6) +6 qtr
Increase in $\sqrt{\text{Fragility}}$	0.010*** (0.003)	0.011*** (0.003)	0.006* (0.003)	0.071*** (0.019)	0.077*** (0.019)	0.049** (0.019)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES
Firm Controls	YES	YES	YES	YES	YES	YES
Adj. R2	0.165	0.153	0.150			
Pseudo R2				0.171	0.150	0.142
Observations	90,701	86,697	81,488	89,987	86,002	80,816
Panel B	Future Overvaluation Indicator			Logit		
	OLS					
	(1) +2 qtr	(2) +4 qtr	(3) +6 qtr	(4) +2 qtr	(5) +4 qtr	(6) +6 qtr
Increase in $\sqrt{\text{Fragility}}$	0.015*** (0.002)	0.012*** (0.003)	0.011*** (0.003)	0.121*** (0.018)	0.096*** (0.019)	0.094*** (0.020)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES
Firm Controls	YES	YES	YES	YES	YES	YES
Adj. R2	0.085	0.077	0.074			
Pseudo R2				0.089	0.079	0.075
Observations	127,692	119,142	111,130	120,087	111,809	104,061

Panel regression of *Undervaluation* (Panel A) and *Overvaluation* (Panel B) on firm-level control variables. *Undervaluation* is a dummy variable that takes the value 1 in the respective quarter for stocks that are in the bottom quartile of the Stambaugh et al. (2015a) measure of misvaluation. Analogously, *Overvaluation* takes the value 1 for stocks that are in the top quartile of the Stambaugh et al. (2015a) measure of misvaluation. The misvaluation measures are evaluated at 2, 4 and 6 quarters ahead (t+2, etc.). Increase in $\sqrt{\text{Fragility}}$ is a dummy variable that takes the value 1 for stocks whose $\sqrt{\text{Fragility}}$ has increased from t-1 until t. The regressions include quarter-year by industry (SIC3) fixed effects with OLS and both quarter-year FE and industry FEs with logit as well as additional (unreported) controls: *Earnings Volatility*, *Ln(Assets)*, *Inst. Ownership*, *Leverage*, *Oper Cash Flow*, *Fixed Assets*, and *Inventory*. Standard errors are adjusted for clustering at the firm level. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 2: fragility and future misvaluation

5.2 Table 3 + Table 4: cash holding and heterogeneous effects

Read: Table 3.

- The benchmark coefficient on $\sqrt{\text{Fragility}}$ is 0.692 and statistically significant.

- Adding controls leaves the coefficient positive and significant.
- The effect remains when multinational firms and the financial crisis are excluded.
- Alternative measures of fragility also point to higher cash holding.

Read: Table 4.

- Fragility matters more during high VIX periods and for firms with high earnings volatility.
- Effects are stronger for financially constrained firms, small firms, firms without bond ratings, and firms with low analyst coverage.
- The product-market fluidity interaction is positive but not statistically significant.

Economic message: Firms build cash when stock prices become more fragile, and the response is strongest where future financing shortfalls are more costly.

Table 3
Stock Fragility and Cash Holding.

	(1) Cash/Assets: All	(2) Cash/Assets: All	(3) Cash/Assets: No MNC	(4) Cash/Assets: No Fin Crisis	(5) Cash/Assets: All	(6) Cash/Assets: All
$\sqrt{Fragility}$	0.692** (0.293)	0.633** (0.250)	0.938*** (0.336)	0.597** (0.275)		
Fragility					39.196** (16.041)	
$\sqrt{Fragility}$ (Current)						0.764*** (0.250)
Earnings volatility	0.086*** (0.019)	0.065*** (0.023)	0.065*** (0.020)	0.085*** (0.018)	0.085*** (0.019)	0.085*** (0.019)
Ln(Assets)	-0.040*** (0.003)	-0.032*** (0.004)	-0.040*** (0.003)	-0.040*** (0.003)	-0.041*** (0.003)	-0.041*** (0.003)
Inst Ownership	0.015*** (0.005)	0.021*** (0.007)	0.017*** (0.005)	0.016*** (0.005)	0.014*** (0.005)	0.014*** (0.005)
Leverage	-0.092*** (0.006)	-0.079*** (0.008)	-0.102*** (0.007)	-0.092*** (0.006)	-0.092*** (0.006)	-0.092*** (0.006)
Fixed Assets	-0.414*** (0.017)	-0.401*** (0.022)	-0.417*** (0.018)	-0.415*** (0.017)	-0.415*** (0.017)	-0.415*** (0.017)
Inventory	-0.550*** (0.024)	-0.515*** (0.033)	-0.548*** (0.025)	-0.550*** (0.025)	-0.552*** (0.025)	-0.552*** (0.025)
Oper Cash Flow	-0.045*** (0.026)	-0.052*** (0.032)	-0.061*** (0.028)	-0.046*** (0.026)	-0.046*** (0.027)	-0.046*** (0.027)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Adj. R2	0.836	0.862	0.882	0.865	0.862	0.863
Observations	136,191	136,191	66,990	119,524	136,449	135,908

Panel regression of cash on Greenwood and Thesmar's stock price fragility measure as well as additional firm-level control variables. The regressions include both firm and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is

Table 3: stock fragility and cash holding

Table 4
Stock Fragility and Cash. The Expected Cost of Misvaluation.

	(1) Cash/Assets	(2) Cash/Assets	(3) Cash/Assets	(4) Cash/Assets	(5) Cash/Assets	(6) Cash/Assets	(7) Cash/Assets
High VIX period $\times \sqrt{Frag.}$	0.916*** (0.240)						
High Earnings Vol. $\times \sqrt{Frag.}$		0.907** (0.428)					
High Fluidity $\times \sqrt{Frag.}$			0.820 (0.567)				
Fin. Constrain $\times \sqrt{Frag.}$				0.802** (0.396)			
Small $\times \sqrt{Frag.}$					1.301*** (0.447)		
No Bond Rating $\times \sqrt{Frag.}$						1.169*** (0.440)	
Low Analyst Cov. $\times \sqrt{Frag.}$							1.305** (0.521)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Adj. R2	0.862	0.862	0.864	0.870	0.862	0.862	0.876
Observations	136,191	136,191	130,473	98,662	136,191	136,191	103,105

Panel regression of cash on Greenwood and Thesmar's stock price fragility measure as well as additional firm-level control variables. The regressions include both firm and quarter-industry (SIC3) fixed effects as well as additional (unreported) controls: *Earnings Volatility*, *Ln(Assets)*, *Inst. Ownership*, *Leverage*, *Oper Cash Flow*, *Fixed Assets*, and *Inventory* as well as the reported interacted variables separately. Standard errors are adjusted for clustering at the firm level. *High VIX Period*, *High Earnings Volatility* and *High (Product Market) Fluidity* are dummy variables that take the value 1 for the top quartile. *Fin. Constrain* is defined as firms above median of Hoeborg and Maksimovic (2015b) measure of Financial Constraints. *Small* is defined as firms in the bottom three quartiles by assets and *No Bond Rating* is a dummy variable that takes the value 1 when a firm lacks a bond rating. *Low Analyst Coverage* is defined as firms in the bottom three quartiles in terms of the number of analysts that follow the firm. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 4: heterogeneity by uncertainty and constraints

5.3 Table 5 + Table 6: expectations and current misvaluation controls

Read: Table 5.

- The cash response to fragility is attenuated when the stock price has recently increased.
- Interactions with positive managerial sentiment are negative, especially in the non-multinational sample.
- This supports the idea that managers' expectations of future undervaluation matter.

Read: Table 6.

- The fragility coefficient stays positive after controlling for current misvaluation proxies.
- The result is robust to book-to-market, residual book-to-market, mutual-fund outflows, and future excess returns.
- The combined-control specification still gives a positive and significant fragility coefficient.

Economic message: The cash result is not simply a contemporaneous valuation effect. Firms respond to the risk of future misvaluation.

Table 5
Managerial Expectations.

	(1) Cash/Assets: All	(2) Cash/Assets: No MNC	(3) Cash/Assets: All	(4) Cash/Assets: No MNC
Stock Price Increase $\times \sqrt{Frag.}$	-0.388*** (0.133)	-0.527*** (0.178)		
Positive Manager Sent. $\times \sqrt{Frag.}$			-0.215 (0.209)	-0.666** (0.287)
Stock Price Increase	0.003*** (0.001)	0.004*** (0.001)		
Positive Manager Sent.			-0.005*** (0.001)	-0.001 (0.002)
$\sqrt{Fragility}$	0.832*** (0.261)	1.198*** (0.349)	0.608* (0.295)	1.204*** (0.396)
Earnings volatility	0.086*** (0.019)	0.065*** (0.023)	0.103*** (0.019)	0.094*** (0.025)
Ln(Assets)	-0.040*** (0.003)	-0.031*** (0.004)	-0.042*** (0.003)	-0.033*** (0.005)
Inst Ownership	0.015*** (0.005)	0.021*** (0.007)	0.017*** (0.005)	0.024*** (0.007)
Leverage	-0.091*** (0.006)	-0.078*** (0.008)	-0.092*** (0.007)	-0.079*** (0.008)
Fixed Assets	-0.414*** (0.017)	-0.402*** (0.022)	-0.415*** (0.018)	-0.398*** (0.022)
Inventory	-0.549*** (0.025)	-0.515*** (0.033)	-0.556*** (0.025)	-0.510*** (0.035)
Oper Cash Flow	-0.046* (0.026)	-0.054* (0.032)	-0.022 (0.028)	-0.026 (0.032)
Quarter-Industry FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Adj. R2	0.862	0.882	0.867	0.887
Observations	136,191	66,990	120,201	56,541

Panel regression of cash on Greenwood and Thesmar's stock price fragility measure interacted with the reported dummy variables as well as additional controls. The regressions include both firm and quarter-industry (SIC3) fixed effects. *Stock Price Increase* is a dummy variable that takes the value 1 for stocks whose price has increased from t-2 until t-1. *Positive Manager Sentiment* is a dummy variable that takes the value 1 if the Jiang-Lee-Marti-Zhou measure of manager sentiment measure was positive in t-1. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 5: managerial expectations

5.4 Table 7 + Table 8: investment, liquidity, and capital structure

Read: Table 7.

- Higher fragility is associated with lower capital expenditures.
- It is also associated with lower R&D and lower repurchases.
- Firms with higher fragility use less short-term debt, consistent with more cautious liquidity management.
- Dividends do not show a robust response.

Read: Table 8.

- There is little evidence that fragility changes equity issuance, long-term debt, or book leverage.
- The response is concentrated in cash, investment, payout flexibility, and short-term debt, not broad capital structure.

Economic message: Firms respond to fragile stock prices by preserving internal liquidity and delaying uses of cash rather than by substantially changing long-term capital structure.

Table 7
Stock Fragility, Investments and Liquidity Management.

	(1) CapEx/Assets	(2) R&D/Assets	(3) Repurch/Assets	(4) Dividends/Assets	(5) ST Debt/Assets
$\sqrt{Fragility}$	-0.050* (0.027)	-0.086* (0.046)	-0.072*** (0.025)	0.007 (0.009)	-0.356*** (0.120)
Earnings volatility	-0.003* (0.002)	-0.009** (0.004)	-0.002*** (0.001)	0.000 (0.000)	-0.000 (0.008)
Ln(Assets)	0.000 (0.000)	-0.009*** (0.001)	0.001*** (0.000)	0.000 (0.000)	-0.006*** (0.001)
Inst Ownership	0.002*** (0.000)	0.001 (0.001)	0.001** (0.000)	-0.000 (0.000)	-0.004 (0.003)
Leverage	-0.018*** (0.001)	-0.004*** (0.001)	-0.011*** (0.000)	-0.003*** (0.000)	0.122*** (0.005)
Fixed Assets	0.004* (0.002)	0.013*** (0.003)	-0.002** (0.001)	-0.001*** (0.000)	0.013* (0.007)
Inventory	0.002 (0.002)	-0.008** (0.004)	-0.004*** (0.001)	-0.002*** (0.001)	0.093*** (0.011)
Oper Cash Flow	0.012*** (0.002)	-0.109*** (0.006)	0.009*** (0.002)	0.003*** (0.001)	-0.057*** (0.010)
Quarter-Industry FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Adj. R2	0.557	0.813	0.276	0.532	0.550
Observations	135,832	73,437	127,068	135,580	135,254

Panel regression of investment and liquidity decisions on Greenwood and Thesmar's stock price fragility measure as well as additional firm-level control variables. The regressions include both firm and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is quarterly from 2001 - 2017. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 7: investment and liquidity management

Table 6
Stock Fragility and Cash. Robust to Proxies for Current Misvaluation.

	(1) Cash/Assets	(2) Cash/Assets	(3) Cash/Assets	(4) Cash/Assets	(5) Cash/Assets	(6) Cash/Assets	(7) Cash/Assets	(8) Cash/Assets
$\sqrt{Fragility}$	0.687*** (0.316)	0.649*** (0.291)	0.706*** (0.295)	0.705*** (0.295)	0.633*** (0.291)	0.628*** (0.325)	0.626*** (0.301)	0.860*** (0.340)
Misprice (SYU)/10	0.002* (0.001)							0.002* (0.001)
Book-to-Market		-0.018*** (0.001)						-0.007*** (0.003)
Residual Book-to-Market (PV)			-0.016*** (0.002)					-0.013*** (0.005)
Residual Book-to-Market (HP)				-0.016*** (0.002)				-0.011*** (0.004)
Residual Book-to-Market (RRV)					-0.009*** (0.001)			-0.020*** (0.003)
Mutual Fund Outflows						0.013*** (0.002)		0.014*** (0.002)
Future Excess Returns/10							-0.001*** (0.000)	-0.009 (0.000)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Adj. R2	0.840	0.842	0.843	0.843	0.842	0.847	0.843	0.851
Observations	94,198	133,332	131,374	131,374	133,167	112,488	121,743	79,692

Panel regression of cash on Greenwood and Thesmar's stock price fragility measure as well as mispricing proxies. Column (1) controls for current mispricing following Stambaugh et al. (2015a) and columns (2)-(8) use a set of mispricing proxies following Derrien et al. (2013) including Book-to-Market measures of Pastor-Veronesi (PV), Hoberg-Philips (HP) and Rhodes-Kropf-Robinson-Viswanathan (RRV). The regressions include both firm and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is quarterly from 2001 - 2017.

Table 6: current misvaluation controls

Table 8
Stock Fragility and Capital Structure.

	(1) Equity Issue (Levels)	(2) Equity Issue (Indicator)	(3) LT Debt/Assets	(4) Leverage (Book)
$\sqrt{Fragility}$	-0.076 (0.079)	-0.184 (0.381)	-0.025 (0.321)	-0.389 (0.343)
Earnings volatility	0.002 (0.009)	-0.054* (0.031)	0.046*** (0.016)	0.081*** (0.018)
Ln(Assets)	-0.017*** (0.001)	-0.075*** (0.001)	0.048*** (0.003)	0.050*** (0.003)
Inst Ownership	0.001 (0.001)	0.016** (0.007)	-0.022*** (0.007)	-0.034*** (0.007)
Fixed Assets	0.015*** (0.004)	0.009 (0.021)	0.134*** (0.018)	0.175*** (0.019)
Inventory	0.010* (0.006)	-0.044 (0.028)	-0.035 (0.026)	0.049* (0.027)
Oper Cash Flow	-0.195*** (0.016)	-0.073 (0.047)	-0.275*** (0.020)	-0.382*** (0.022)
Quarter-Industry FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Adj. R2	0.175	0.170	0.744	0.758
Observations	128,046	128,046	139,175	139,175

Panel regression of equity issuance, leverage and long term debt on Greenwood and Thesmar's stock price fragility measure as well as additional firm-level control variables. Equity issuance in levels scaled by assets, and indicator variable of equity issuance equal to 1 if equity increase by at least 1%, otherwise set to 0. The regressions include both firm and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is quarterly from 2001 - 2017. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 8: capital structure

5.5 Figure 1 + Table 9: BlackRock–BGI event-study evidence

Read: Figure 1.

- The upper panel shows treated firms increase cash after the BlackRock–BGI merger.
- The lower panel shows the merger also increases fund-family-level fragility.
- There is little evidence of differential pre-trends before the merger announcement and completion.

Read: Table 9.

- The BlackRock–BGI merger raises return volatility and excess-return volatility.
- Smaller bank/asset-management mergers also raise excess-return volatility in some specifications.
- The volatility evidence supports the claim that ownership shocks increased stock price fragility.

Economic message: The event-study and volatility evidence strengthen identification by showing that an ownership-concentration shock raised fragility and then corporate cash responded.

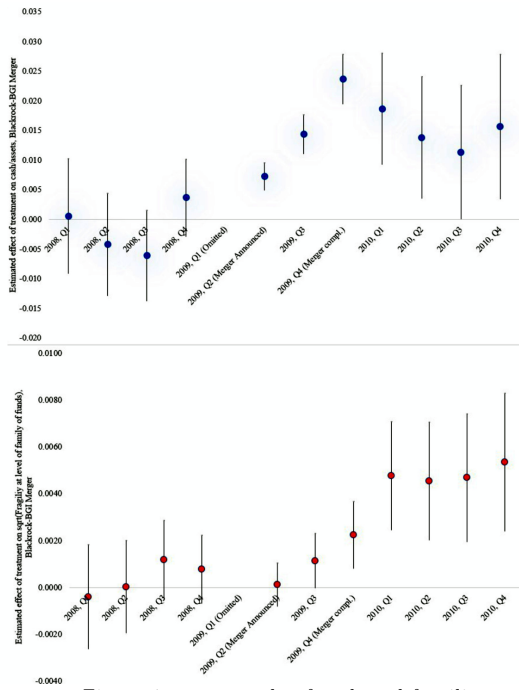


Figure 1: event study of cash and fragility

Table 9
Mergers and Volatility.

	(1) Vol. of Returns	(2) Vol. of Excess Returns	(3) Vol. of Returns	(4) Vol. of Excess Returns
BlackRock-BGI Merger	0.004*** (0.001)	0.007*** (0.001)		
BofA-Fleet, JPM-Bank One Mergers			0.000 (0.001)	0.002*** (0.001)
Earnings volatility	0.009** (0.005)	0.009* (0.005)	0.019*** (0.007)	0.020*** (0.006)
Ln(Assets)	-0.006*** (0.001)	-0.006*** (0.001)	-0.001 (0.001)	-0.001 (0.001)
Inst Ownership	0.001 (0.002)	-0.000 (0.002)	0.002 (0.002)	0.001 (0.002)
Leverage	0.043*** (0.004)	0.043*** (0.004)	0.024*** (0.003)	0.025*** (0.003)
Fixed Assets	0.005 (0.005)	0.007 (0.005)	0.014*** (0.005)	0.015*** (0.005)
Inventory	0.008 (0.008)	0.006 (0.007)	-0.002 (0.007)	-0.001 (0.007)
Oper Cash Flow	-0.029*** (0.008)	-0.032*** (0.008)	-0.026*** (0.008)	-0.026*** (0.008)
Quarter-Industry FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Adj. R2	0.619	0.604	0.640	0.652
Observations	23,775	23,775	23,310	23,310

Panel regression of volatility of (excess) returns on merger treatment indicators as well as additional firm-level control variables. The regressions include both firm and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. In Columns (1)-(2) the data is quarterly from 2008Q1-2010Q4 and Merger Treatment equals one for firms treated by the Blackrock and BGI merger after its completion. In Columns (3)-(4) the data is quarterly from Q3 2002 until Q3 2005 and Merger Treatment equals one for firms treated by the Bank of America-Fleet and JP Morgan-Bank One mergers for the period after the merger. Statistical significance at the 1, 5, or 10% levels is reported as ***, **, *, respectively.

Table 9: mergers and volatility

5.6 Table 10 + Table 11: direct BlackRock–BGI treatment effects

Read: Table 10.

- The BlackRock–BGI treatment raises cash/assets by about 0.013 in the baseline.
- The estimate is stable across sample exclusions and product-market competition robustness checks.
- The result implies a meaningful increase relative to average cash holdings.

Read: Table 11.

- The merger treatment lowers capital expenditures, R&D, repurchases, dividends, and short-term debt.
- These signs are consistent with cautious liquidity management after an exogenous increase in fragility.

Economic message: The natural experiment reproduces the panel-regression pattern: more fragility leads to more cash and less cash outflow.

Table 10
Effect of Exogenous Shock to Fragility (BlackRock-BGI Merger) on Cash Holding.

	All	Sample Robustness			Competition Robustness		
	(1) Cash/Assets: All	(2) Cash/Assets: Exclude Select Industries	(3) Cash/Assets: Exclude Low B/M	(4) Cash/Assets: Exclude High R&D	(5) Cash/Assets: All	(6) Cash/Assets: Exclude Largest Increase in P.M. Concentration	(7) Cash/Assets: Exclude More Treated Industries
Merger Treatment	0.013*** (0.002)	0.013*** (0.002)	0.012*** (0.002)	0.014*** (0.003)	0.013*** (0.002)	0.013*** (0.002)	0.013*** (0.003)
Prod. Market HH/100					-0.008 (0.553)		
Earnings volatility	0.201*** (0.028)	0.174*** (0.033)	0.227*** (0.033)	0.216*** (0.043)	0.198*** (0.028)	0.200*** (0.029)	0.196*** (0.039)
Ln(Assets)	-0.029*** (0.007)	-0.037*** (0.007)	-0.043*** (0.010)	-0.042*** (0.012)	-0.029*** (0.007)	-0.029*** (0.010)	-0.031*** (0.010)
Inst Ownership	-0.003 (0.009)	-0.000 (0.009)	-0.007 (0.011)	-0.012 (0.013)	-0.004 (0.009)	-0.010 (0.009)	-0.011 (0.016)
Leverage	-0.024*** (0.008)	-0.021*** (0.007)	-0.022*** (0.008)	-0.020*** (0.013)	-0.023*** (0.008)	-0.027*** (0.008)	-0.019 (0.014)
Fixed Assets	-0.404*** (0.033)	-0.376*** (0.034)	-0.330*** (0.036)	-0.381*** (0.068)	-0.404*** (0.033)	-0.395*** (0.033)	-0.525*** (0.054)
Inventory	-0.438*** (0.035)	-0.449*** (0.035)	-0.423*** (0.039)	-0.370*** (0.060)	-0.438*** (0.033)	-0.450*** (0.035)	-0.489*** (0.062)
Oper Cash Flow	0.098*** (0.022)	0.069*** (0.022)	0.073*** (0.026)	0.086*** (0.053)	0.095*** (0.022)	0.107*** (0.025)	0.136*** (0.038)
Quarter-Industry FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Adj. R2	0.930	0.913	0.929	0.901	0.930	0.932	0.935
Observations	22,178	18,581	10,832	6,311	21,868	18,655	9,933

Panel regression of cash on BlackRock-BGI treatment indicator as well as additional firm-level control variables. The regressions include firm fixed effects.

Table 10: BlackRock-BGI shock and cash

tests in the 30, 50 and 100 percentiles of the merger, statistical significance at the 1, 5, or 10% levels is reported as ***, **, *

Table 11
Effect of Exogenous Shock to Fragility (BlackRock-BGI Merger).

	(1) CapEx/Assets	(2) R&D/Assets	(3) Repurch/Assets	(4) Dividends/Assets	(5) ST Debt/Assets
Merger Treatment	-0.002*** (0.000)	-0.001** (0.000)	-0.002*** (0.000)	-0.000** (0.000)	-0.005*** (0.001)
Earnings volatility	-0.005 (0.004)	-0.017* (0.009)	0.003 (0.002)	0.001 (0.001)	0.018 (0.016)
Ln(Assets)	-0.002 (0.001)	-0.013*** (0.003)	0.003*** (0.001)	0.001*** (0.000)	-0.001 (0.003)
Inst Ownership	0.003* (0.001)	0.001 (0.002)	-0.002 (0.002)	0.000 (0.000)	0.001 (0.006)
Leverage	-0.017*** (0.002)	-0.000 (0.002)	-0.010*** (0.001)	-0.002*** (0.000)	0.084*** (0.008)
Fixed Assets	-0.033*** (0.007)	-0.004 (0.011)	-0.014*** (0.003)	-0.004*** (0.001)	-0.040*** (0.016)
Inventory	0.009* (0.004)	-0.005 (0.007)	-0.012*** (0.003)	-0.001 (0.001)	0.089*** (0.024)
Oper Cash Flow	0.013*** (0.004)	-0.054*** (0.012)	-0.000 (0.002)	0.000 (0.001)	-0.063*** (0.017)
Quarter-Industry FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES
Adj. R2	0.619	0.881	0.316	0.639	0.648
Observations	22,148	11,976	20,931	22,127	21,995

Panel regression of investment and liquidity management decisions on BlackRock-BGI treatment indicator as well as additional firm-level control variables. The regressions include firm fixed effects and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is quarterly from 2008Q1-2010Q4. Merger Treatment equals one

Table 11: BlackRock-BGI shock and other policies

5.7 Table 12: alternative asset-management merger shocks

Table 12
Effect of Alternative Shocks (BoA-Fleet and JP Morgan-Bank One Mergers).

	(1) Cash/Assets	(2) CapEx/Assets	(3) Cash/Assets	(4) CapEx/Assets
Merger Treatment (both mergers)	0.005*** (0.002)	-0.000 (0.000)		
Merger Treatment (BoA-Fleet)			0.007*** (0.002)	-0.000 (0.000)
Earnings volatility	0.053* (0.028)	-0.009*** (0.003)	0.052 (0.032)	-0.008** (0.003)
Ln(Assets)	-0.035*** (0.008)	0.000 (0.001)	-0.045*** (0.009)	-0.000 (0.001)
Inst Ownership	0.013 (0.010)	0.005*** (0.001)	0.015 (0.012)	0.006*** (0.001)
Leverage	-0.062*** (0.010)	-0.013*** (0.001)	-0.061*** (0.012)	-0.013*** (0.001)
Fixed Assets	-0.411*** (0.031)	-0.016*** (0.004)	-0.392*** (0.037)	-0.015*** (0.004)
Inventory	-0.410*** (0.040)	0.006* (0.003)	-0.443*** (0.048)	0.009** (0.004)
Oper Cash Flow	0.048 (0.034)	-0.002 (0.003)	0.004 (0.042)	-0.002 (0.003)
Quarter-Industry FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Adj. R2	0.931	0.611	0.928	0.636
Observations	23,842	23,754	18,473	18,430

Panel regression of cash holding and investment for merger treatment as well as additional firm-level control variables. The regressions include firm fixed effects and quarter-industry (SIC3) fixed effects and the standard errors are adjusted for clustering at the firm level. The data is quarterly from Q3 2002 until Q3 2005. In Columns 1 and 2 Merger Treatment takes

Table 12: alternative shocks from BoA-Fleet and JP Morgan-Bank One mergers

Read:

- Combining the BoA-Fleet and JP Morgan-Bank One mergers gives a positive cash effect.
- The BoA-Fleet treatment alone gives a larger positive cash coefficient.
- Investment effects are not statistically meaningful in these smaller merger events.

Economic message: The smaller mergers provide additional support for the cash-response mechanism, though their real-investment effects are weaker than in the BlackRock-BGI event.

6 Short summary

- The paper studies whether firms respond to stock price fragility, defined as exposure to non-fundamental demand shocks from ownership structure.
- The model predicts that higher fragility raises the expected cost of future external finance when the firm may be underpriced.

- Firms should therefore hold more precautionary cash and reduce investment or other cash uses.
- Fragility is measured using the Greenwood–Thesmar ownership-flow covariance measure.
- Increases in fragility predict future undervaluation and overvaluation indicators.
- Panel regressions show that higher fragility raises cash holdings.
- The effect is stronger when uncertainty and financing constraints are higher.
- The cash result survives controls for current misvaluation proxies.
- Fragility is associated with lower investment, R&D, repurchases, and short-term debt.
- It is not strongly associated with changes in capital structure.
- The BlackRock–BGI merger provides a large exogenous shock to ownership concentration and fragility.
- Treated firms increase cash and reduce investment/liquidity outflows after the merger.
- Alternative asset-management mergers provide additional evidence, especially for cash holdings.

7 Conclusion

7.1 Core conclusion

Friberg, Goldstein, and Hankins show that managers view stock price fragility as a real financial risk. When mutual-fund ownership patterns make a stock more exposed to non-fundamental demand shocks, firms respond ex ante by increasing cash and reducing investment or liquidity outflows.

7.2 Economic intuition

The key intuition is precautionary. A firm may need to raise external funds in the future. If its stock price is underpriced at that time, issuing equity or relying on market-based financing is costly. Higher fragility makes such misvaluation more likely, so the firm holds more cash today to reduce the chance of being forced into expensive financing later.

7.3 Takeaway

The paper adds a new dimension to the real effects of financial markets. Non-fundamental stock-market shocks do not only matter after prices move. They also shape managers' expectations and precautionary corporate policies before shocks occur.